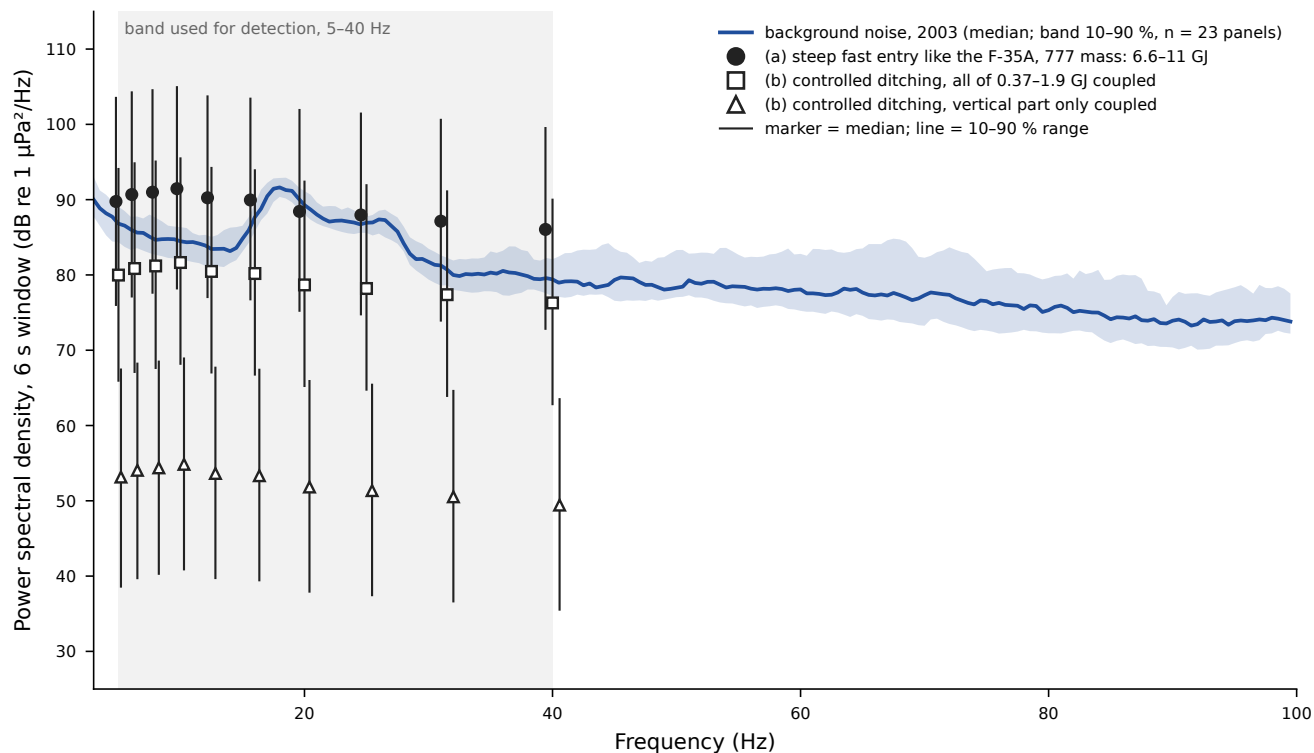


Cape Leeuwin (H01W): background noise against the sound expected from two kinds of impact



What the chart shows: the background noise at Cape Leeuwin (H01W) (curve), and the sound level that an impact near 37.2°S 89.6°E, 2,221 km away, would make there (markers, one set per frequency band). (a) is a steep, fast entry like the F-35A crash, with the mass of a 777. (b) is a controlled ditching. Where a marker is above the curve, the impact would stand out in that band. We do not know how well a ditching makes sound, so (b) has two limits. Estimates only: the sound level per joule comes from one crash, and the noise is from 2003.

Technical: received energy spectral density = $\eta \cdot E_{pc} / 2\pi \cdot S_b(t) \cdot 10^{-(TL_b/10)} \cdot 10^{-(C/10)}$, shown as PSD over a 6 s window (Blackman's convention) in third-octave bands; η = F-35A $\eta_{cal}(\tau, z_s)$, RAM-corrected, paired with $\tau \log-U(0.05, 10)$ s, $z_s \in \{2, 10, 30\}$ m, F^{-2} roll-off; $C = N(0, 10 \text{ dB})$ site + $N(0, 3 \text{ dB})$ receiver. TL: KRACKEN – RAM Δ on the median impact path (stand-in impact-PDF median, 37.23°S 89.58°E; 2,221 km). $m = 175$ t; (a) $v U(270, 360)$ m/s; (b) $v U(60, 150)$ m/s, flight-path angle $U(1.4^\circ, 4^\circ)$, vertical KE 0.5-6 MJ. $n = 40,000$ draws. Noise: Blackman 2004 App. B (prereg e137521). EXPLORATORY; not a detection P_D .